Here, I' is the intensity of polarized light existing between P_3 and P_2. $I' = \frac{I}{4}$.
Substituting,

$$I'' = \left(\frac{I}{4}\right)\cos^2(45^\circ) = \left(\frac{I}{4}\right)\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{I}{8}$$

$$I'' = \frac{I}{8}$$

7.5.3.4. Uses of polaroids

1. Polaroids are used in goggles and cameras to avoid glare of light.
2. Polaroids are used to take 3D pictures i.e., holography.
3. Polaroids are used to improve contrast in old oil paintings.
4. Polaroids are used in optical stress analysis.
5. Polaroids are used as window glasses to control the intensity of incoming light.
6. Polarised laser beam acts as needle to read/write in compact discs (CDs).
7. Polarised light is used in liquid crystal display (LCD).

7.5.4 Polarisation by reflection

The simplest method of producing plane polarised light is by reflection. Consider a beam of unpolarised light incident on a polished glass surface *XY*. This light undergoes reflection as well as refraction. As it is unpolarized, it consists of vibrations which are parallel to the reflecting surface (shown as dots) and also not parallel to it (shown as arrows). It is shown in Figure 7.35. For a particular angle of incidence, the reflected light is found to be plane polarised and the refracted light is found to be partially polarised. It is because, the parallel vibrations to the surface are reflected and the other vibrations are refracted. Few parallel vibrations may also get refracted resulting in partially polarised refracted light. **The angle of incidence for which the reflected light is found to be plane polarised is called *polarising angle* i_p.**

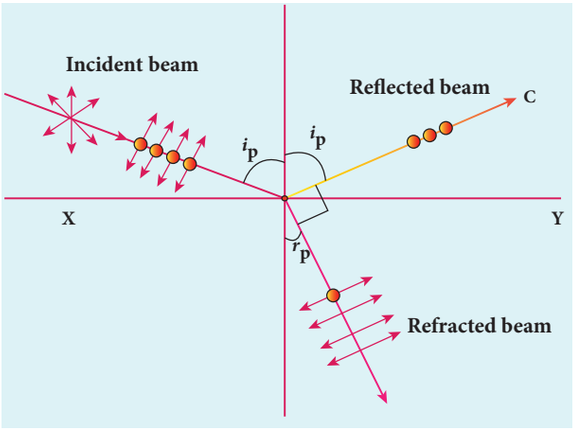


Figure 7.31 Polarisation by reflection

7.5.4.1 Brewster's Law

The British Physicist, Sir. David Brewster found that at the polarising angle, the reflected and the refracted rays are perpendicular to each other. Suppose i_p is the polarising angle and r_p is the angle of refraction, from the geometry as shown in Figure 7.35, we can write,

$$r_p = 90^0 - i_p \qquad (7.56)$$

From Snell's law, the refractive index n of the medium with respect to air is,

$$\frac{\sin i_p}{\sin r_p} = n \qquad (7.57)$$

Substituting equation (7.56) in (7.57), we get,

$$\frac{\sin i_p}{\sin\left(90^\circ - i_p\right)} = \frac{\sin i_p}{\cos i_p} = n$$

$$\tan i_p = n \qquad (7.58)$$

UNIT-7(XII-Physics_Vol-2)_new.indd 84 21-07-2020 16:36:21